University of Colorado **Boulder**

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ASEN 5335: Aerospace Environment

Module 5: The Radiation Environment

Xinlin Li

The Core

The Core

- **Which of the following statements are correct?**
  - Everybody on Earth is dead in a year
  - Invisible field of energy made up of ~~electricity~~ and magnetism
  - Called the electromagnetic field
  - Protects us from cosmic radiation
  - The field is falling apart
  - Aaron Eckhart's description of the Earth's core
  - Physics 101, hot metal moving fast drives EM field
  - Planes will fall from the sky
  - Electronics will be fried
  - Static discharges in the atmosphere will lead to superstorms
  - EM field shields us from the solar winds, a lethal blend of radioactive particles and microwaves
  - Microwave radiation will literally cook our planet

What really shields us from the solar wind etc?

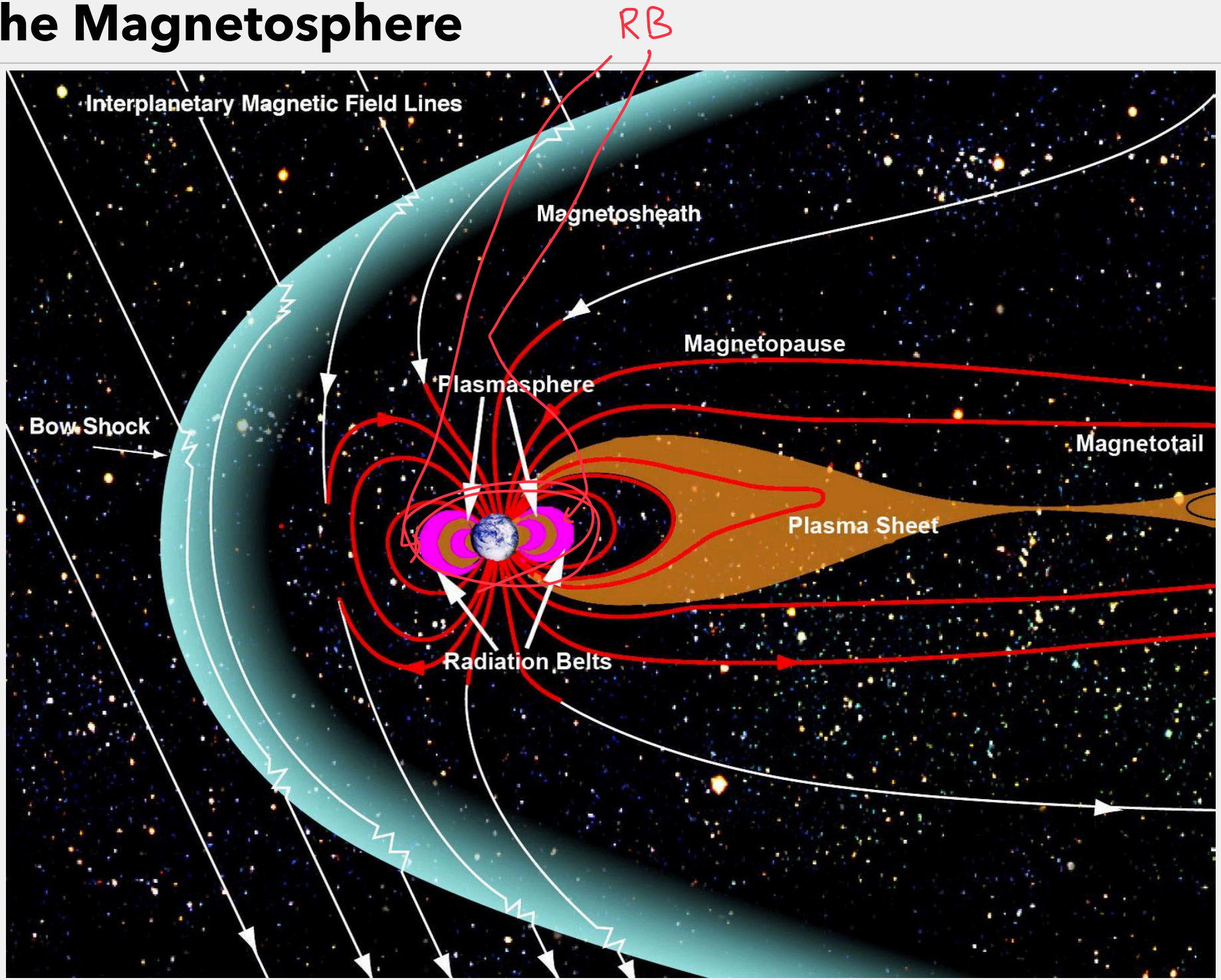
- Solar wind itself is probably not that harmful;
  it requires > 100 MeV protons to reach the ground
- Solar SEPs, X-rays, gamma-rays are “radiation”
- There is nothing “radioactive”

- **What would happen if B-field disappeared?**

- **Where do X-rays, SEPs, etc get absorbed?**

→ atmosphere

The Magnetosphere



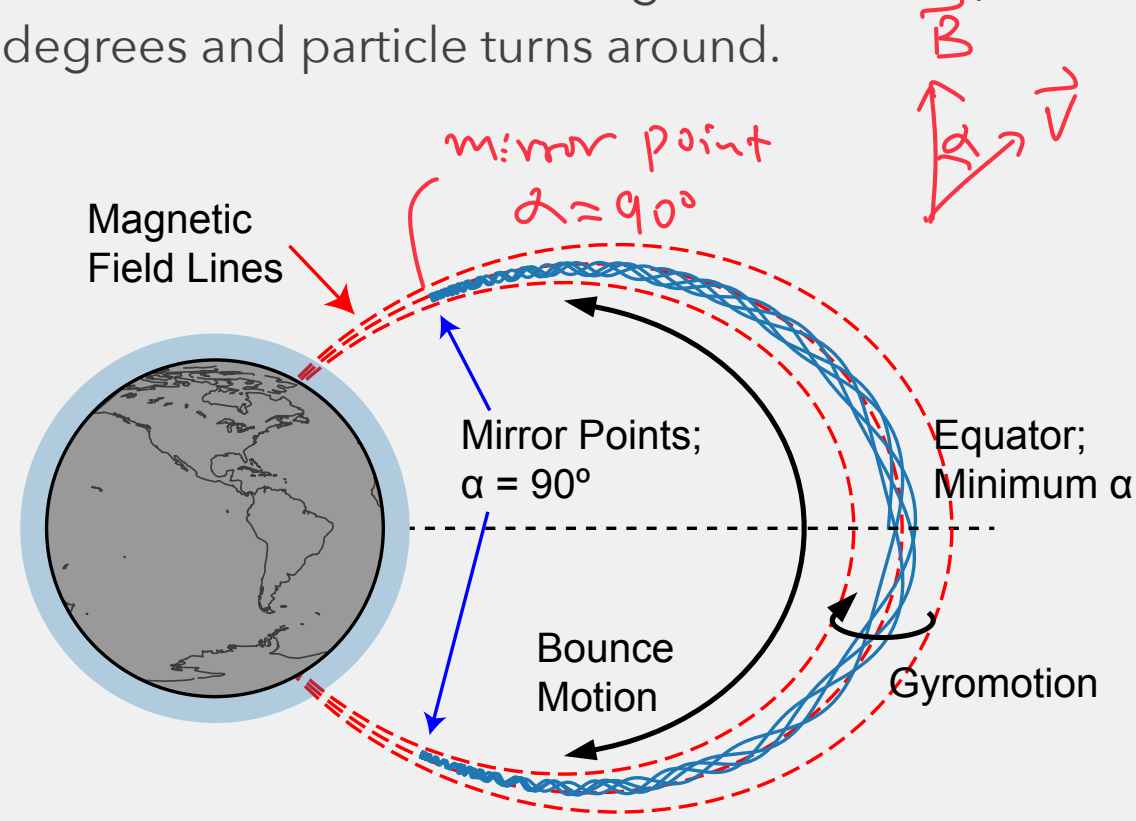
Goals of This Module

- **Understand basic structure and dynamics of the Radiation Belts**
  - Origin; particles and energies; locations
  - Dynamics of radiation belts: solar cycle, geomagnetic storms, enhancements and losses
- **Understand radiation associated with SEPs and GCRs**
- **Understand effects of energetic radiation on spacecraft**
  - Total dose effects
  - Single event effects, deep dielectric discharge, etc.
- **Learn about Radiation measurements and science of Van Allen Probes, CSSWE, and CIRBE**
- **Understand how the Apollo missions dealt with the Radiation Belts**
  - And how future missions will need to as well

Review of Particle Trapping

- Remember: particles "bounce" in the magnetic bottle (or mirror) with converging magnetic field lines (increasing intensity) at either end
- Results simply from Lorentz force, **F** = q (**v** x **B**)
- Pitch angle increases in increasing B due to conservation of magnetic moment; at some point pitch angle goes to 90 degrees and particle turns around.

$$\mu = \frac{\frac{1}{2} m v_\perp^2}{B} = \frac{\frac{1}{2} m v^2 \sin^2\alpha}{B}$$



Review of Gradient Drift

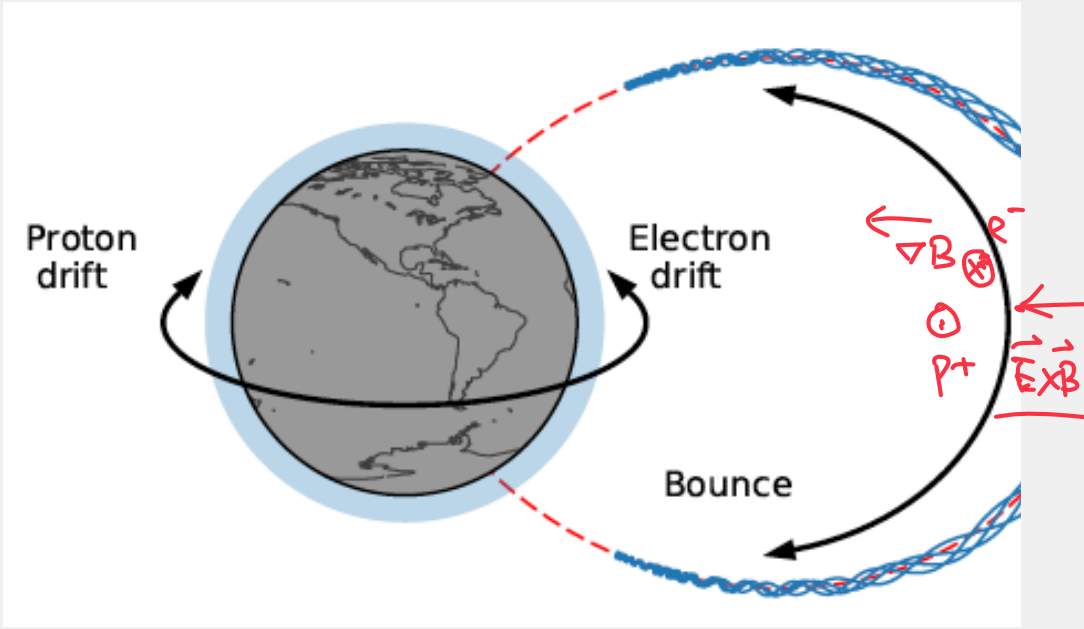
$$\vec{F} = -\mu \nabla B = -\frac{\frac{1}{2} m v_\perp^2}{B} \nabla B$$

$$\vec{v}_d = \frac{\vec{F} \times \vec{B}}{qB^2} = \frac{m v_\perp^2}{2q} \frac{\vec{B} \times \nabla B}{B^3}$$

$$= E_{KL} \frac{\vec{B} \times \nabla B}{B^2} \propto \frac{1}{L^2}$$

$$t_d = \frac{d}{v_d} \sim \frac{2\pi r}{v_d}, \quad r = R_L$$

$$= \propto \frac{1}{E_K L}$$



$L =$	1.5	3	6
1 keV	488 hours	244 h	122 h
100 keV	4.88 h	2.44 h	1.22 h
1 MeV	29.3 min	14.6 min	7.3 min
100 MeV	17.6 sec	8.8 sec	4.4 sec

t_d

Effects of Relativity

rest → $E = m_0 c^2$

$E_{rest} \rightarrow m_0 \begin{cases} e^-: & 511\,keV \\ p^+: & 938\,MeV \end{cases}$

$E = mc^2 = \gamma m_0 c^2$

$E_K = (\gamma - 1) m_0 c^2$

$\gamma = \dfrac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$, $\quad v = c\sqrt{1 - \dfrac{1}{\left(\frac{E_K}{m_0 c^2} + 1\right)^2}}$

at $v = \frac{c}{2}$: $\quad E_{ke} = 79\,keV$

$E_p \simeq 145\,MeV$

relativity matters: $e^- > 100\,keV$

$p^+ > 100\,MeV$

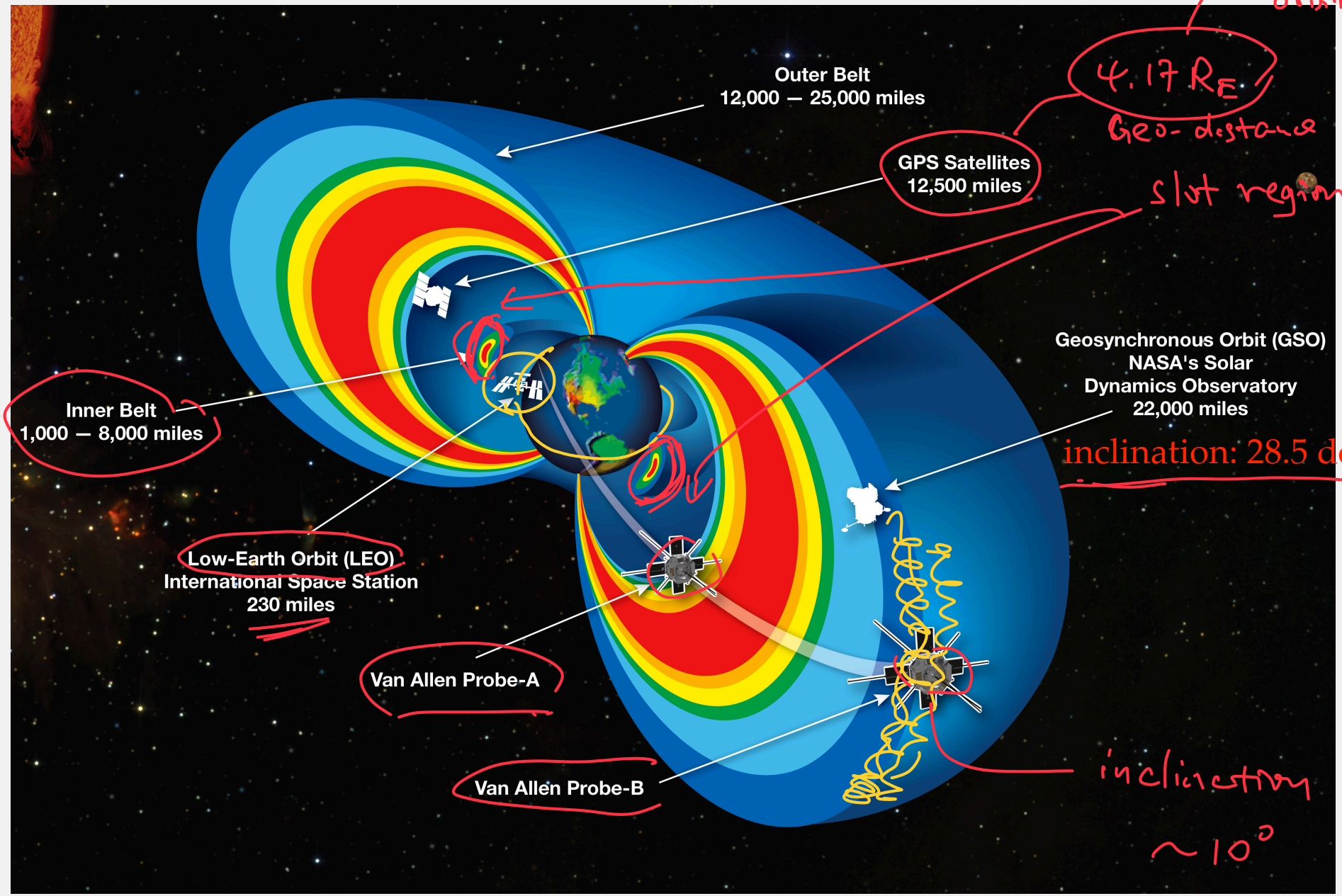
$M = \gamma m_0$

$r_g = \dfrac{m v_\perp}{qB} = \dfrac{\gamma m_0 v_\perp}{qB}$

$\omega_c = \dfrac{qB}{m} \Rightarrow \dfrac{qB}{\gamma m_0}$

$\gamma > 1$

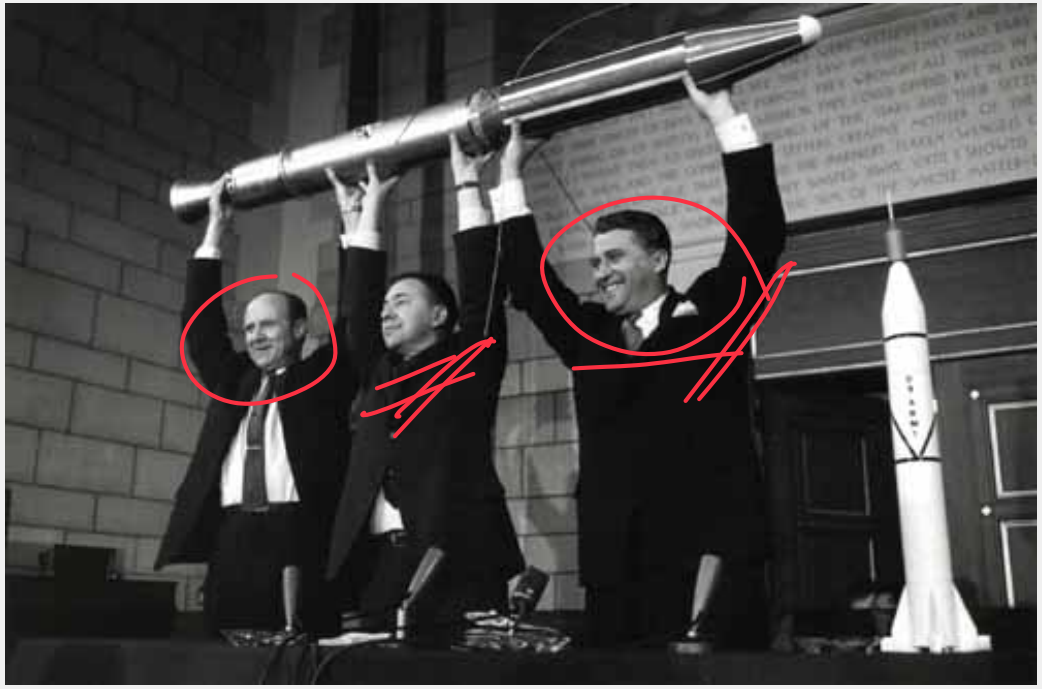
The Radiation Belts



https://upload.wikimedia.org/wikipedia/commons/b/b4/Comparison_satellite_navigation_orbits.svg

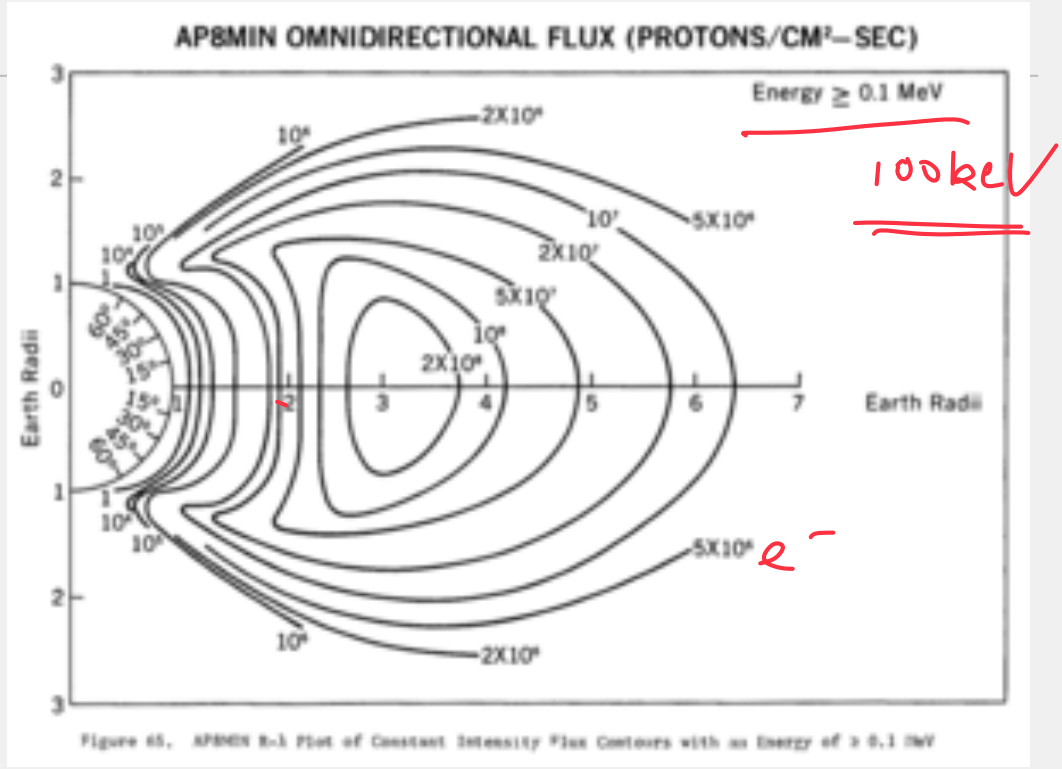
Radiation Belt Discovery

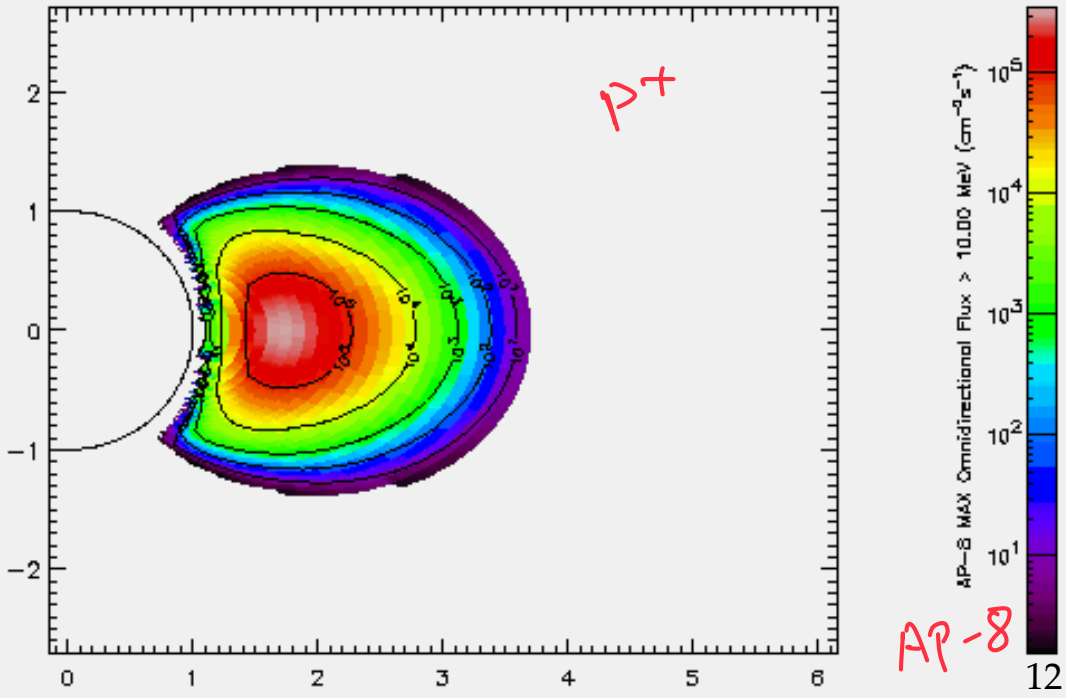
- 1958: Explorer 1: First American satellite
- Had a radiation detector (Geiger counter) and a micrometeoroid detector
- Radiation trapped in magnetic fields was expected, but the high persistent radiation levels were not!



Inner Belt

- Extends from L (~700 km)
  - There are plenty of 100s of keV (up to 1.5 MeV) electrons
- Mostly protons with energies from 1 MeV to 100's of MeV (some to GeV)
  - recall: 145 MeV corresponds to v = c/2
- Very stable populations of protons
  - Why? inner magnetosphere not significantly perturbed by space weather effects; protons too heavy to scatter with EM waves





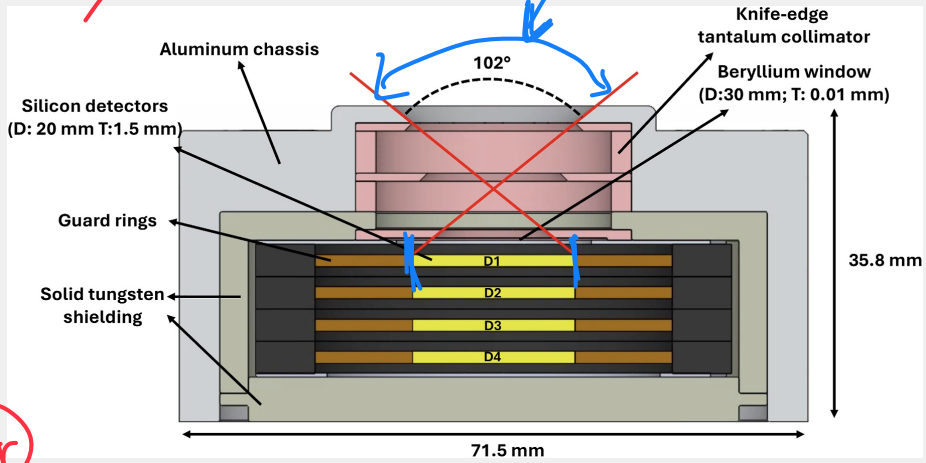
Density vs. Flux

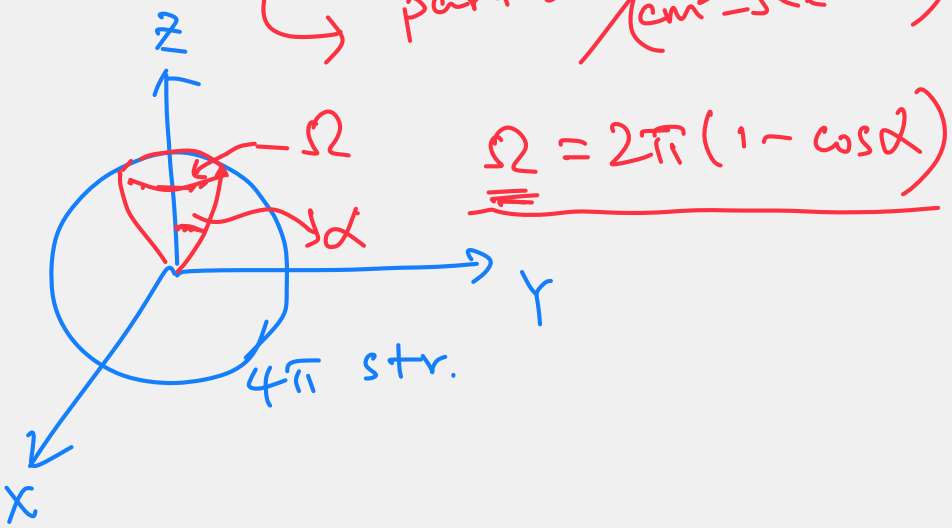
- Why do we use "flux" (electrons/m^2/sec) instead of density (electrons/m^3) to describe radiation belt populations?

collect: N/sec particles/sec

per unit area: $\frac{N}{A \cdot sec}$: particles/cm^2/sec

per unit angle & per unit area & per sec: $\frac{N}{A\Omega\, sec}$: particles/(cm^2-sec-str)



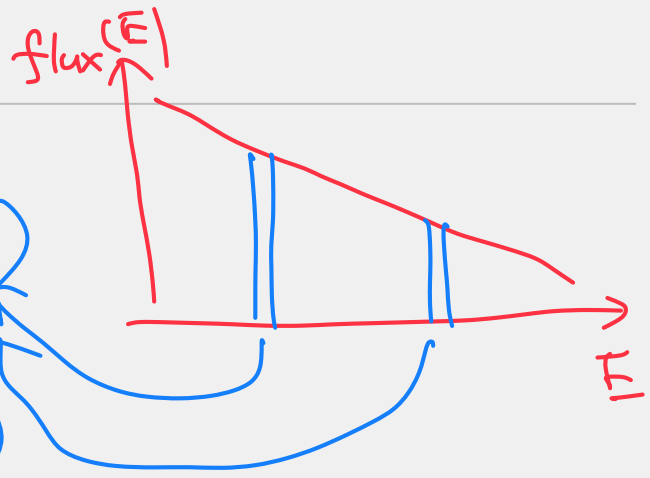


$\Omega = 2\pi(1 - \cos\alpha)$

$A\Omega$ – geometric factor

cm^2 · str.

Different Flux Units



- **Differential** directional flux

$F(\alpha, E) \Rightarrow$ particles/(cm²-str-s-kev) $\Rightarrow$ #/(cm²-str-s-kev)

pitch angle, Energy

- ***Integral*** directional flux

$$F(\alpha) = \int_E F(\alpha, E)\, dE \rightarrow \#/(cm^2\text{-str-sec})$$

- **Differential** omnidirectional flux

$$F(E) = \int_\Omega F(\alpha, E)\, d\Omega \rightarrow \#/(cm^2\text{-sec-kev})$$

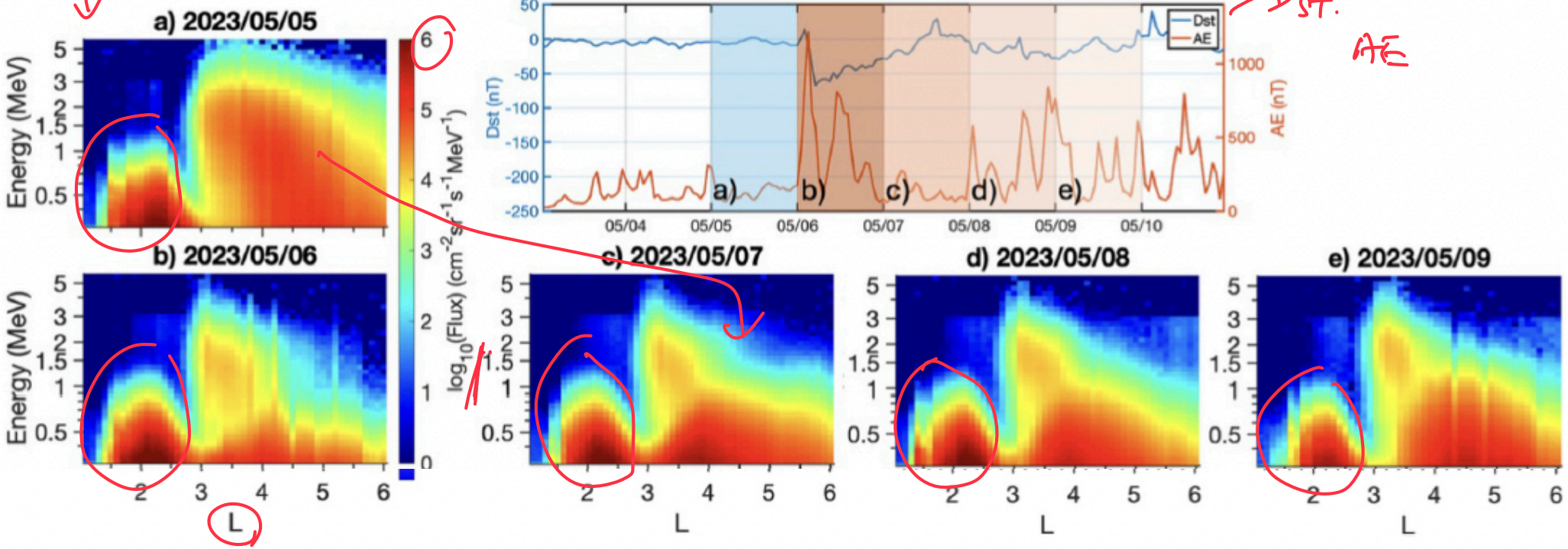
- ***Integral*** omnidirectional flux

$$F = \int_E \int_\Omega F(\alpha, E)\, d\Omega\, dE \rightarrow \#/(cm^2\text{-sec})$$

Inner Belt Electrons

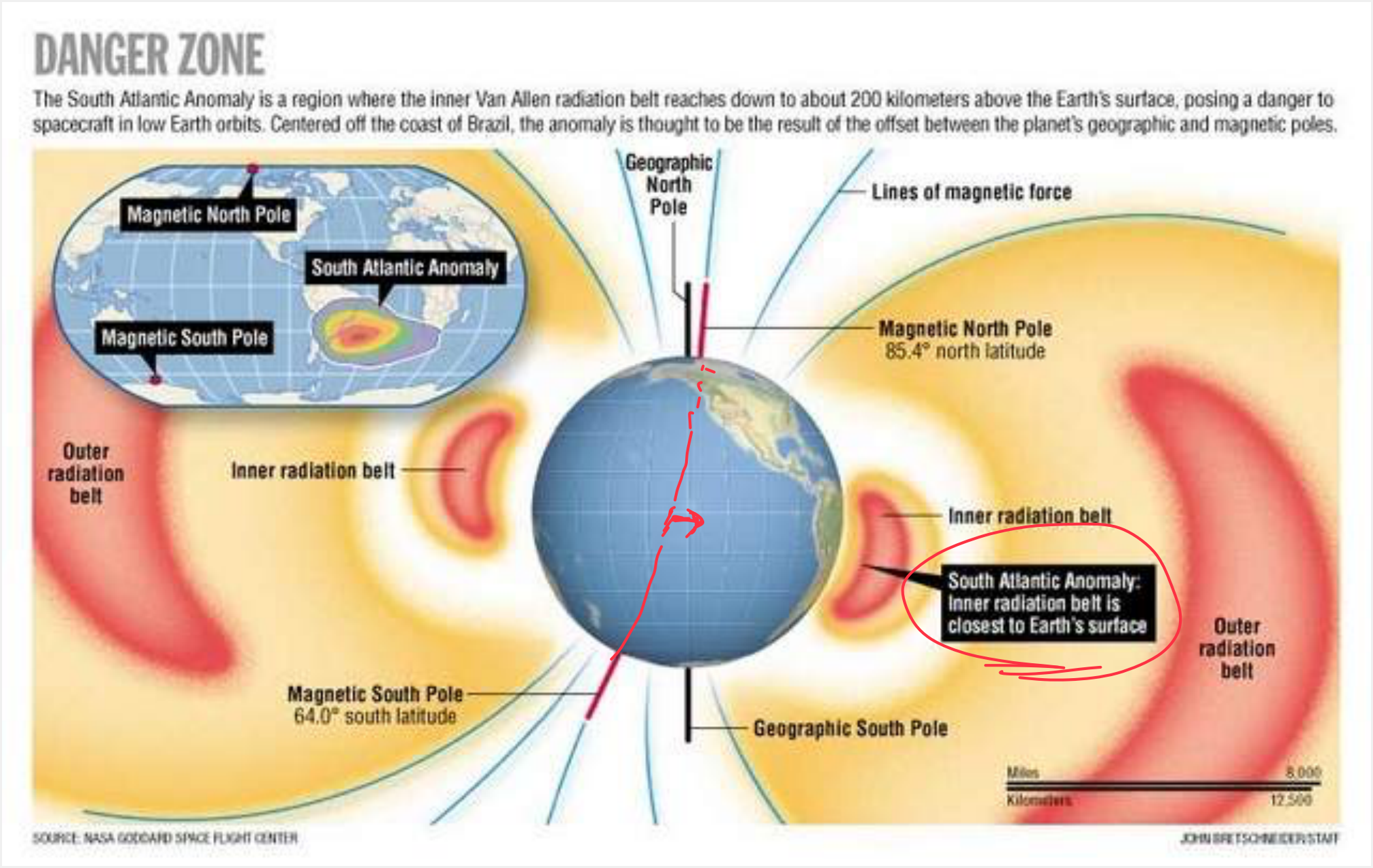
- Electrons mostly < 1 MeV
- Fluxes vary significantly with time: not nearly as stable as protons

CIRBE/REPTile-2 Measurements

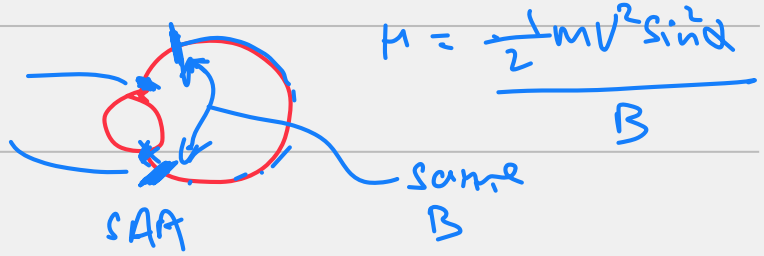


(Chien, Misty, X. Li, JGR, 2026)

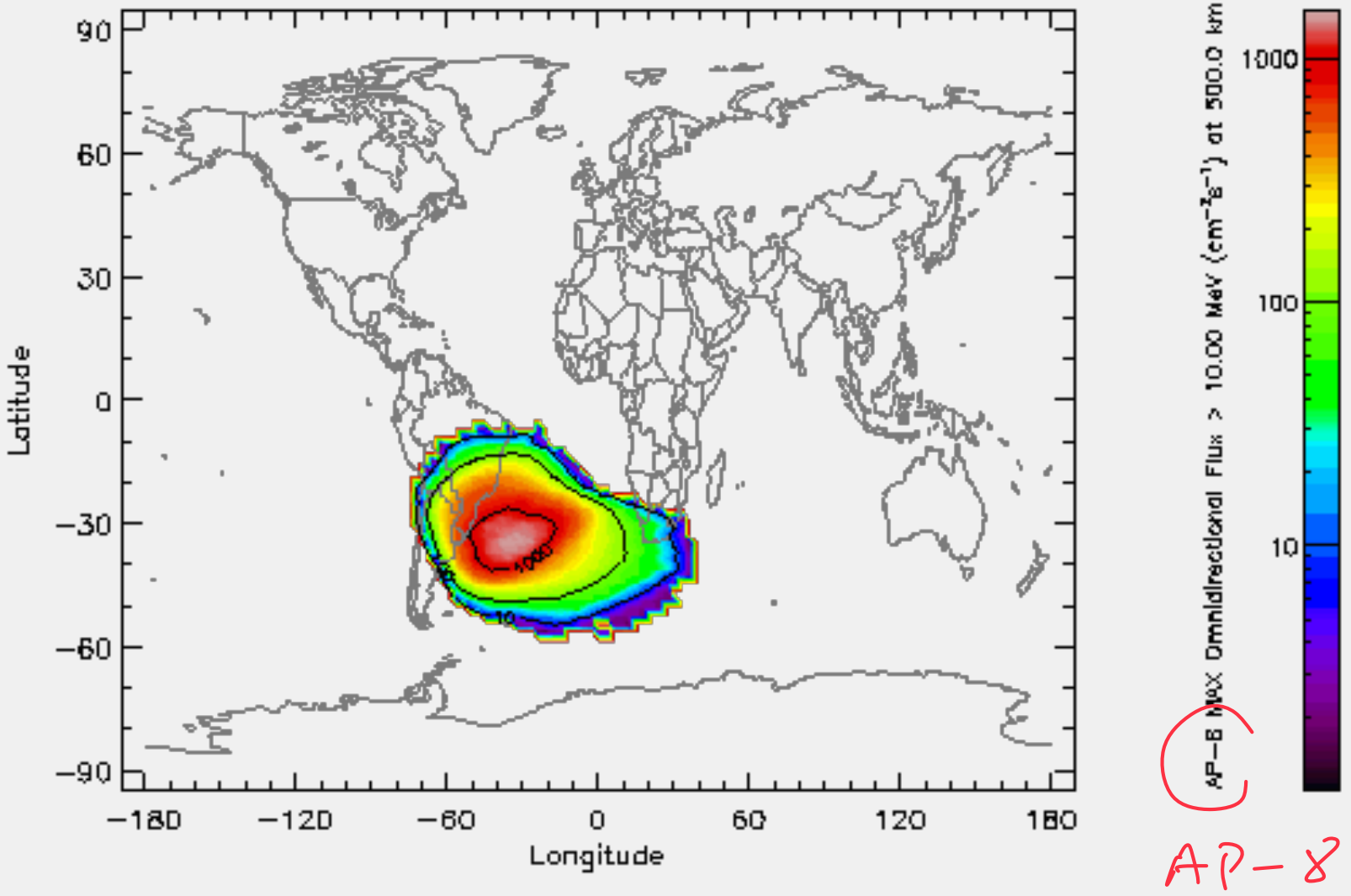
The South Atlantic Anomaly



South Atlantic Anomaly

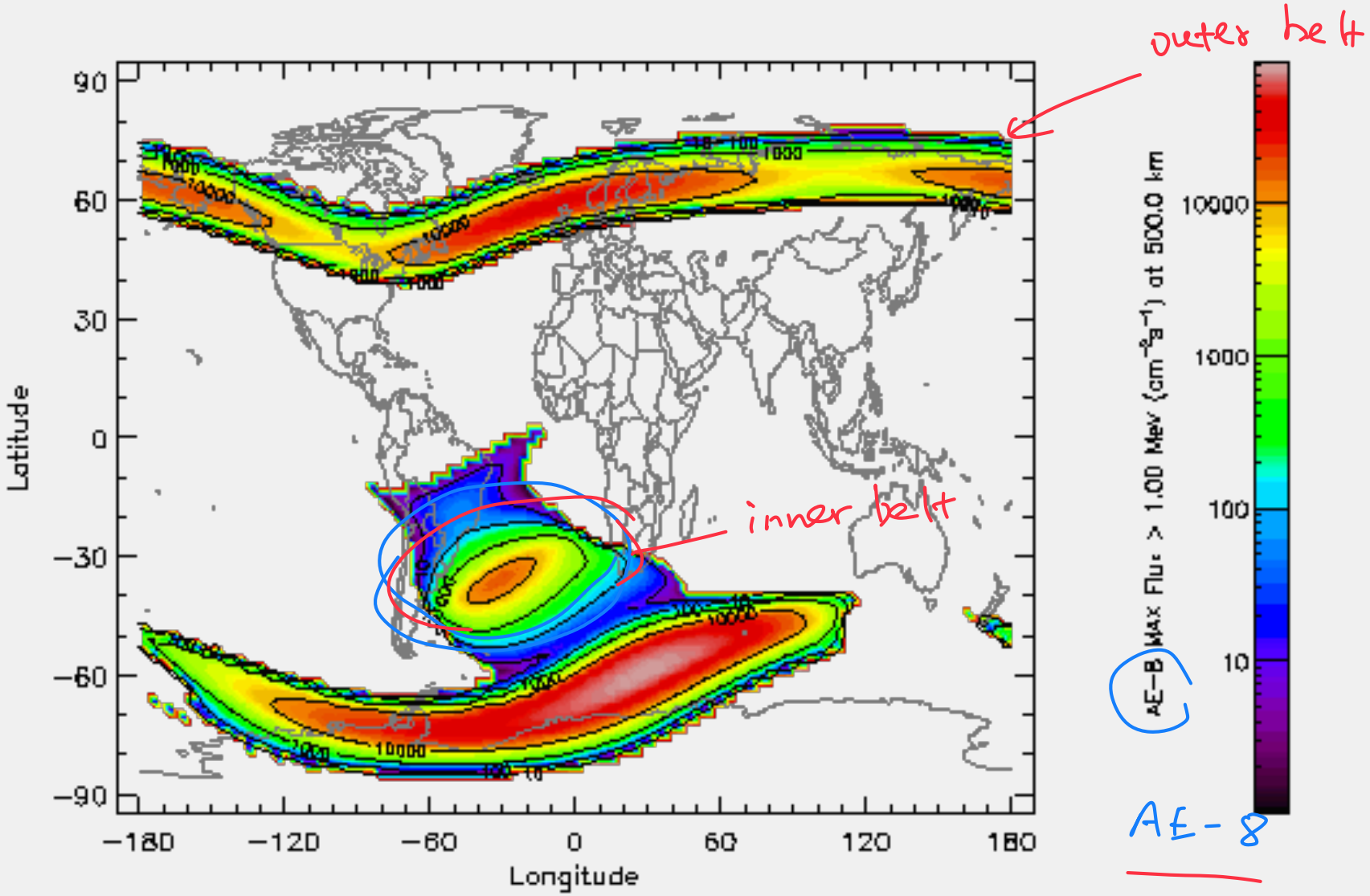


- Proton Fluxes, energies > 10 MeV, at 500 km altitude; up to ~2000 protons/cm²/sec



SAA Electrons

- Evidence that there are electrons in the inner radiation belt!
- Plot shows electron fluxes > 1 MeV at 500 km altitude; peak of ~20,000 el/cm$_2$/sec

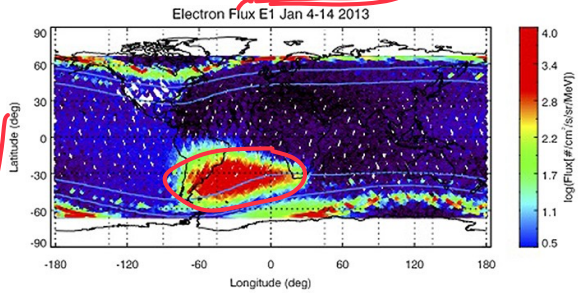
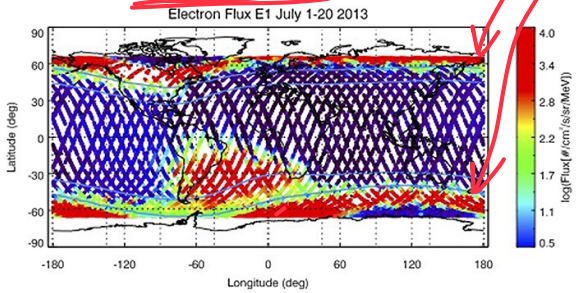
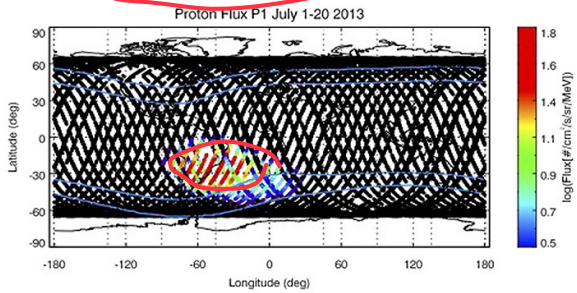
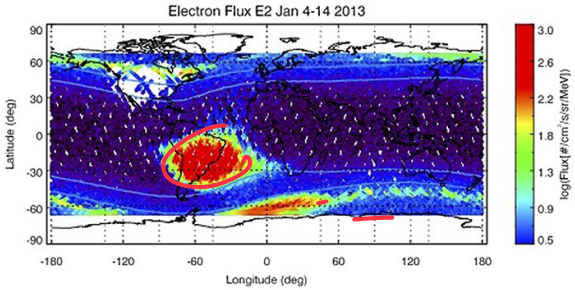
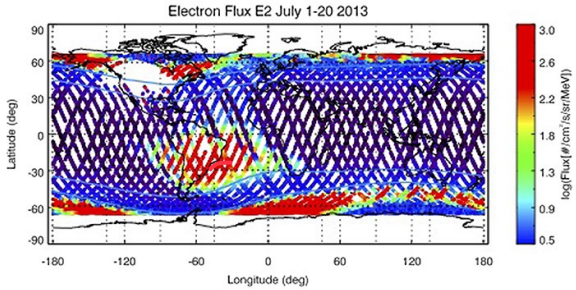
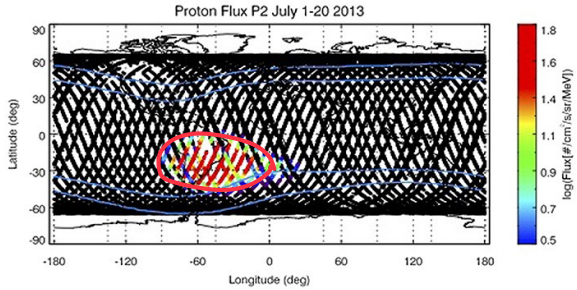
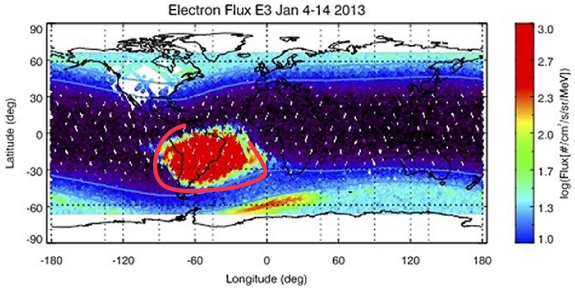
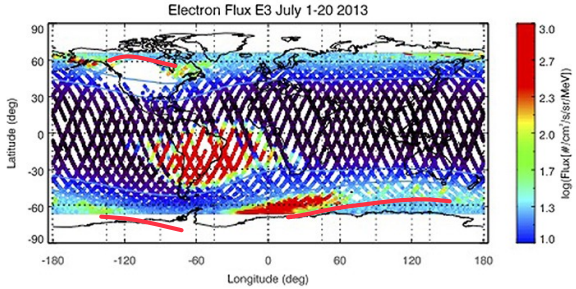
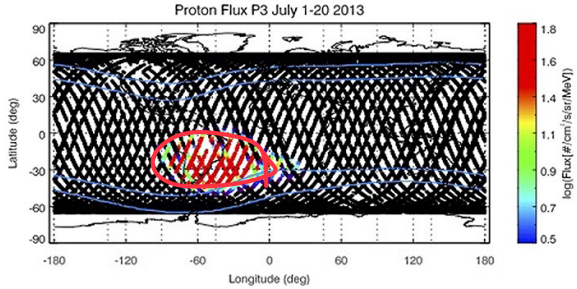


Electrons and Protons measured by CSSWE/REPTile

(inner belt and outer belt)

Observations by CSSWE (CubeSat)

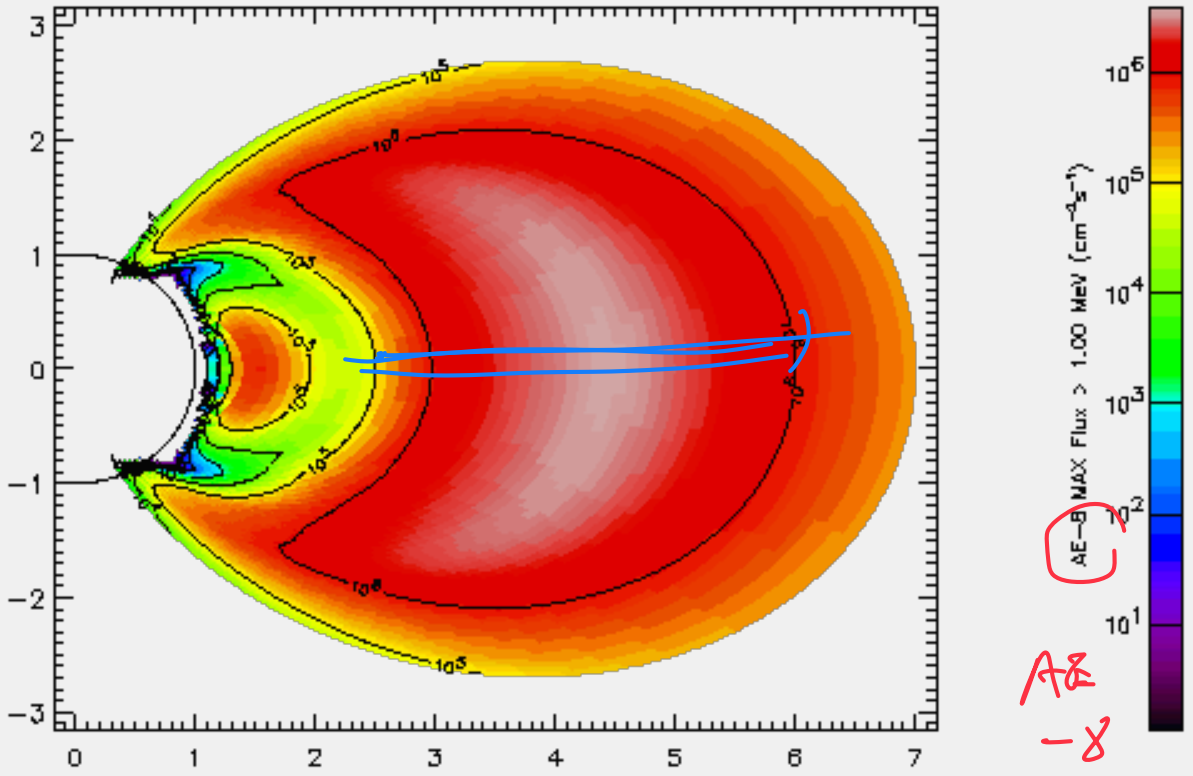
outer belt

	Quiet/Electron	Active/Electron	Active/Proton
Channel 1 E: 0.68-1.63 MeV P: 9-18 MeV			
Channel 2 E: 1.63-3.8 MeV P: 18-30 MeV			
Channel 3 E: >3.8 MeV P: 30-40 MeV			

(Zhang, Li, et al., JGR, 2017)

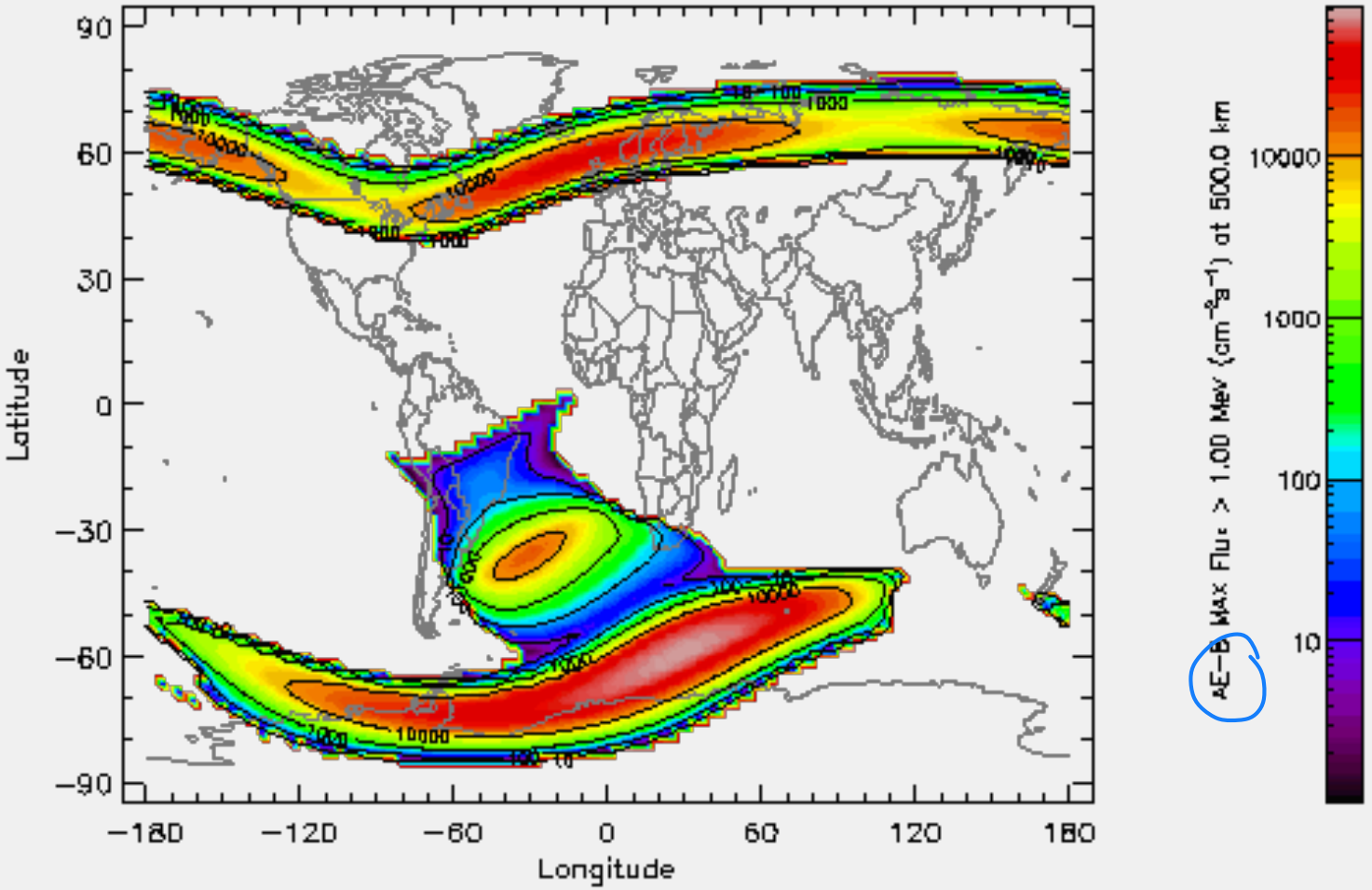
Outer Belt

- Primarily electrons from 100 keV to 10 MeV
  - Note speeds: at 100 keV, v ~ 0.55c; at 10 MeV, v ~0.999c
- Densities peak at about 10^6–10^7 electrons/cm^2/sec (above 1 MeV)
- Bounded by L-shells between ~3 and 6–7
- Fluxes extend to high latitudes compared to inner belt; significant flux at low-Earth orbit at ~50-70 degree latitudes



Outer Belt in LEO

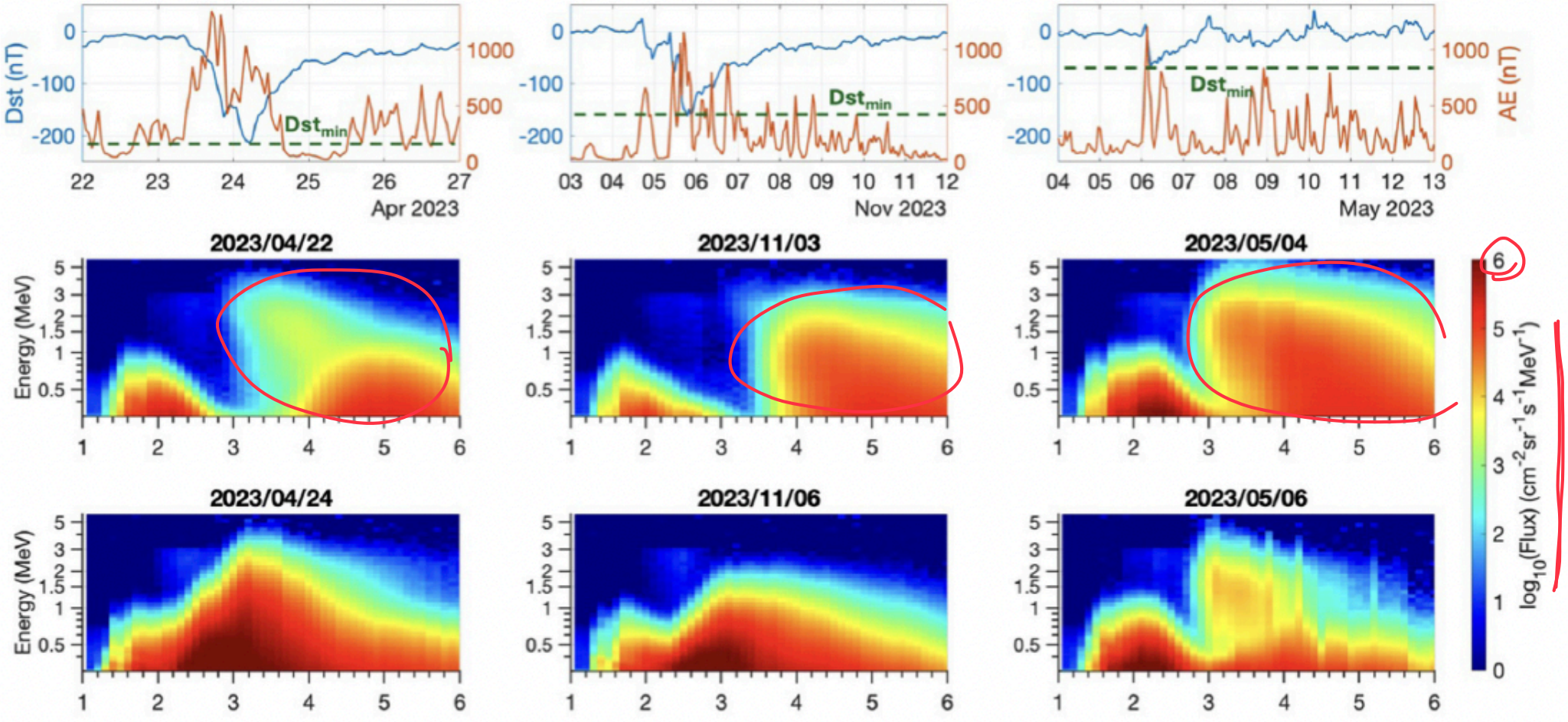
- But, compared to the heart of the outer belt, fluxes are 100x lower



Slot Region

- **Why is there this region devoid of particles? Ideas:**
  - Electromagnetic waves scatter them into the atmosphere at low L-shells
  - Inward diffusion is somehow limited by some physical process

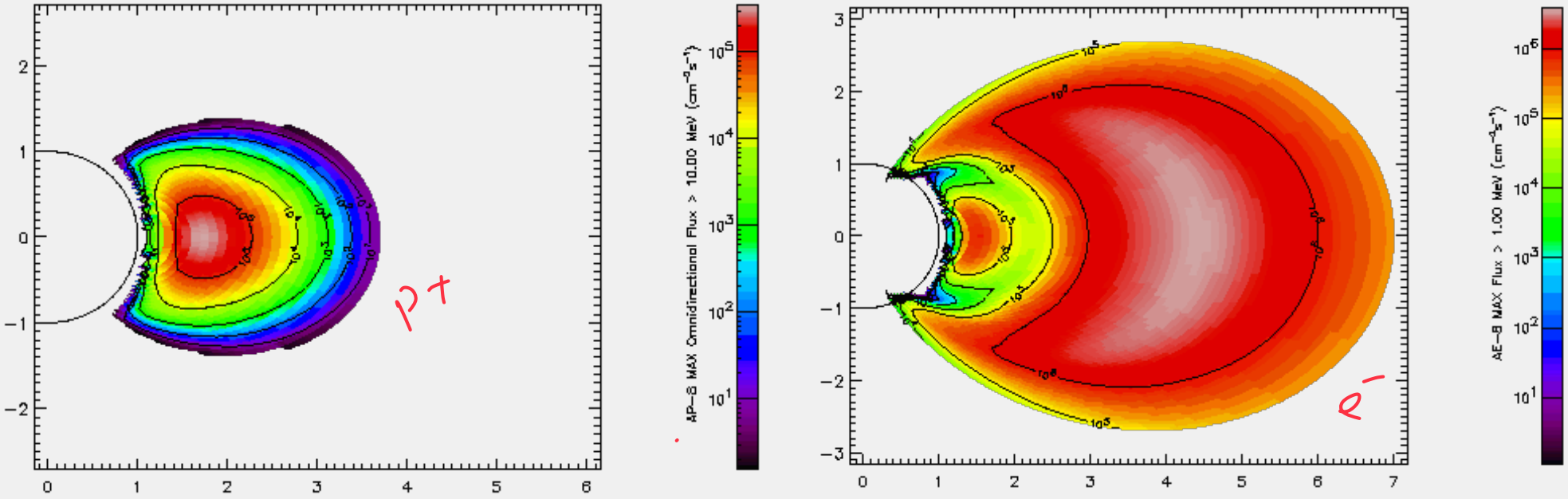
CIRBE/REPTile-2 Measurements



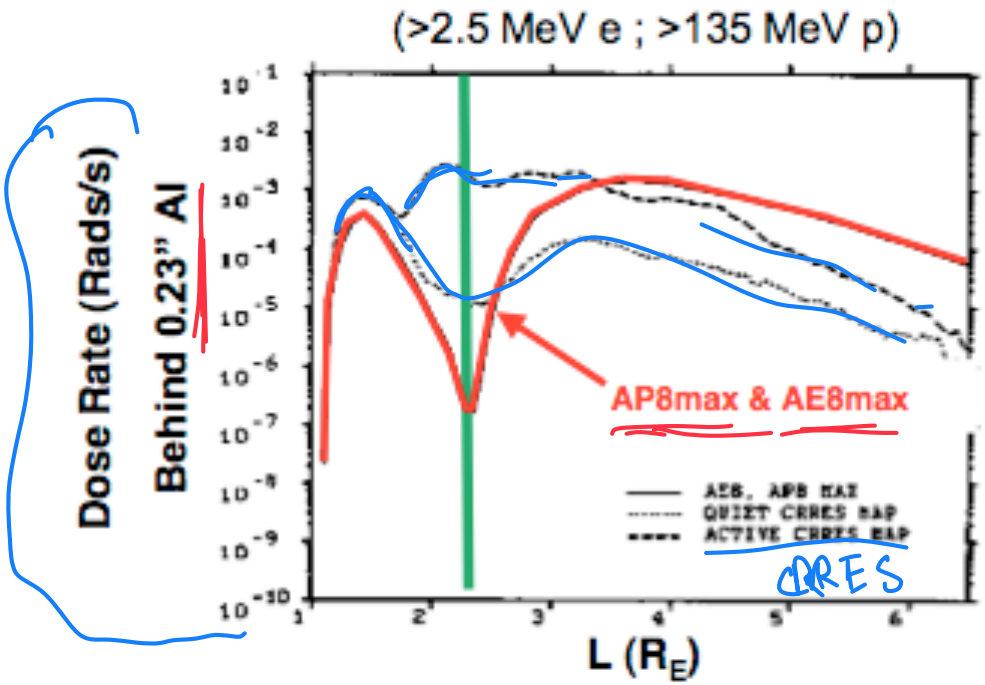
(Chien, Misty, X. Li, JGR, 2026)

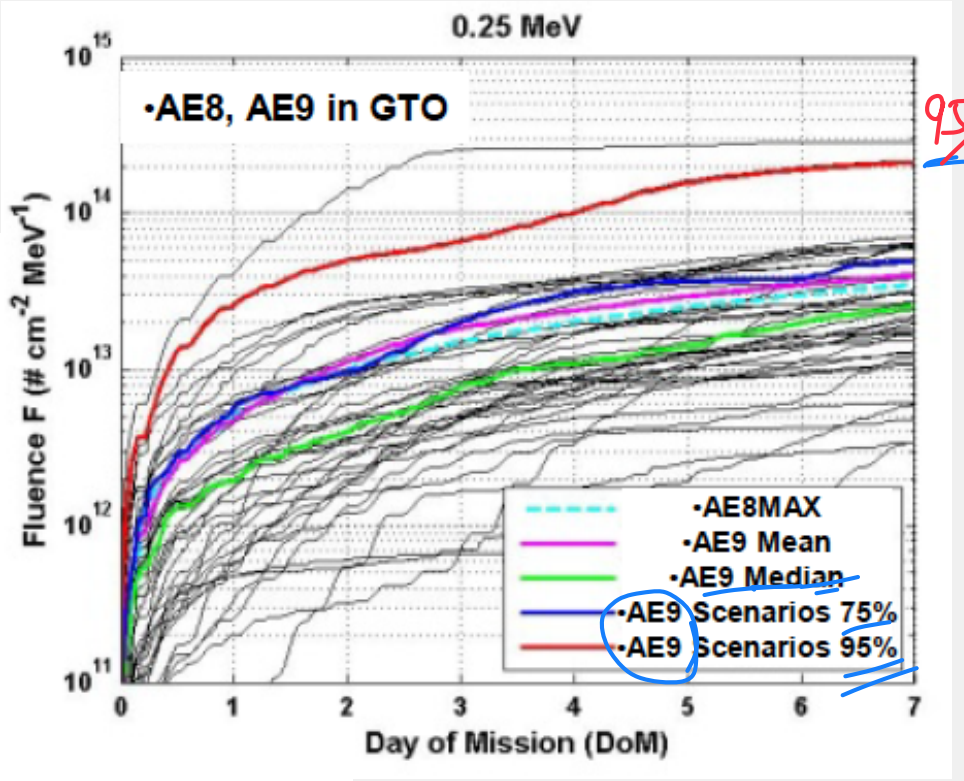
Radiation Belt Models: AE8 and AP8

- Say you want to do some radiation belt modeling. You want an estimate of the fluxes in the radiation belts: both electrons and protons
- AE8: electrons
- AP8: protons
- MAX and MIN conditions: AE8MAX, AE8MIN, AP8MAX, AP8MIN
- https://ccmc.gsfc.nasa.gov/modelweb/models/trap.php



AE8 and AP8 models are not very good



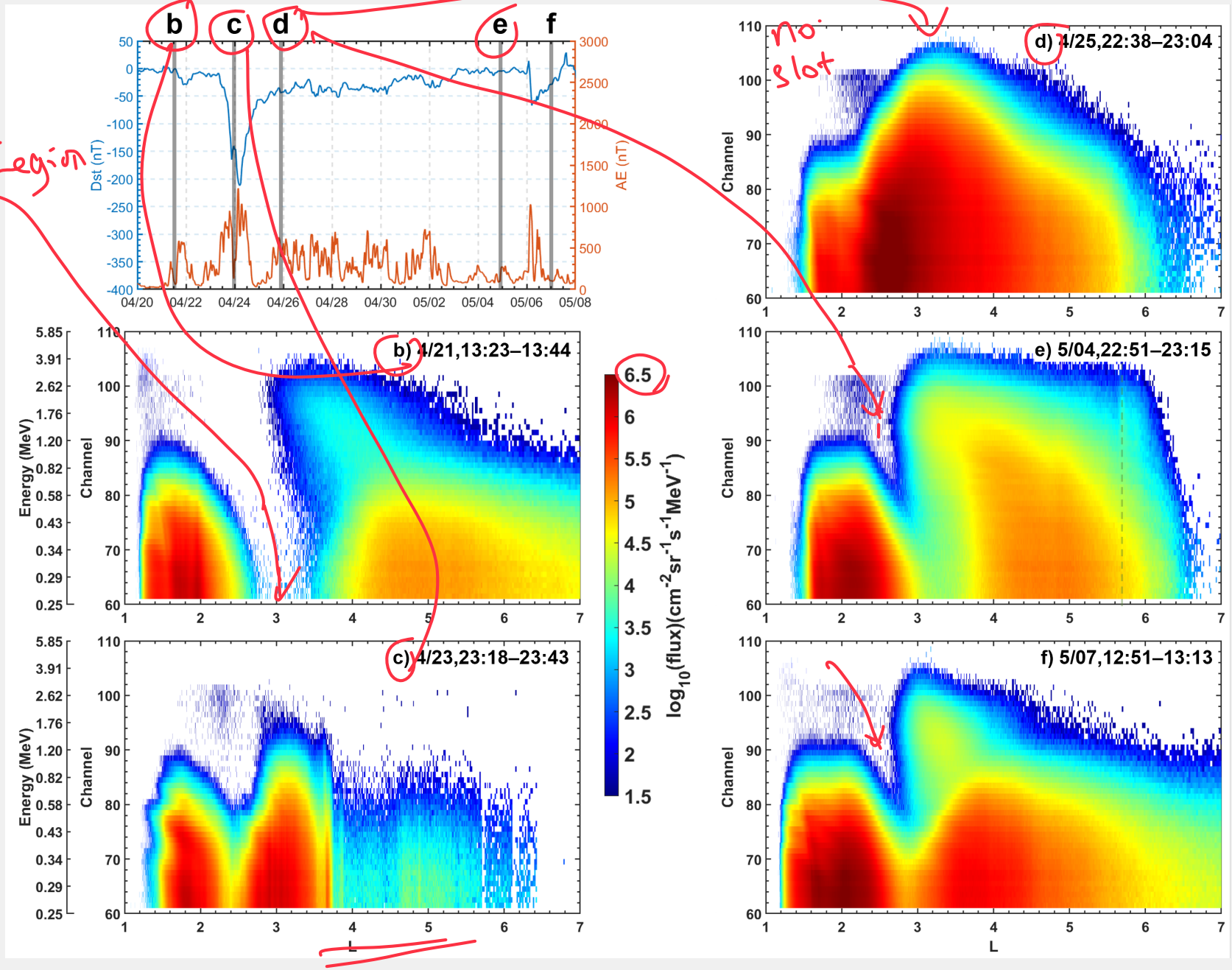


For MEO orbit (L=2.2), #years to reach 100 kRad:

- Quiet conditions (NASA AP8, AE8) : 88 yrs
- Active conditions (CRRES active) : 1.1 yrs

AE8 & AP8 under estimate the dose for 0.23" shielding

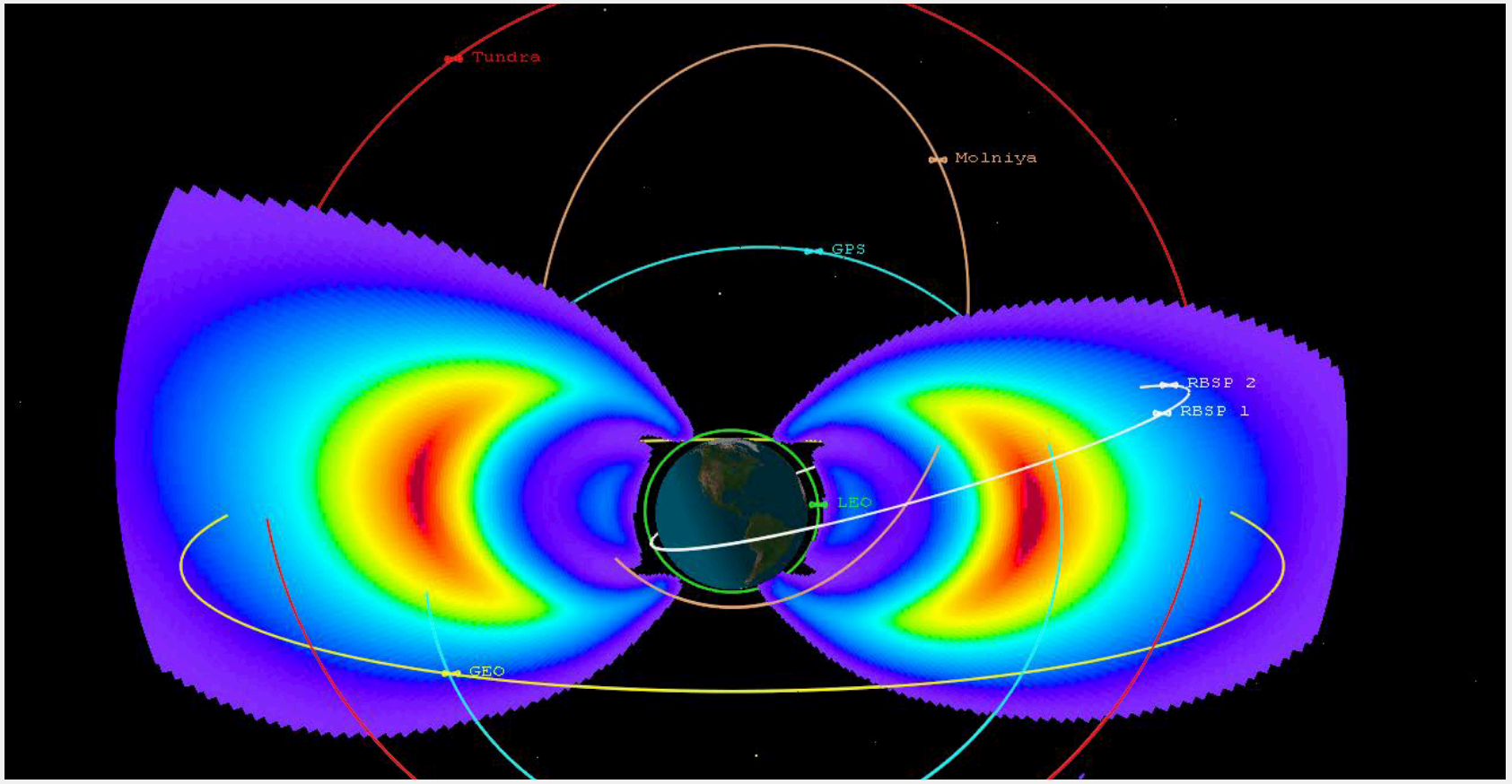
First Results From REPTile-2 Measurements Onboard CIRBE



(Li et al., GRL, 2024)

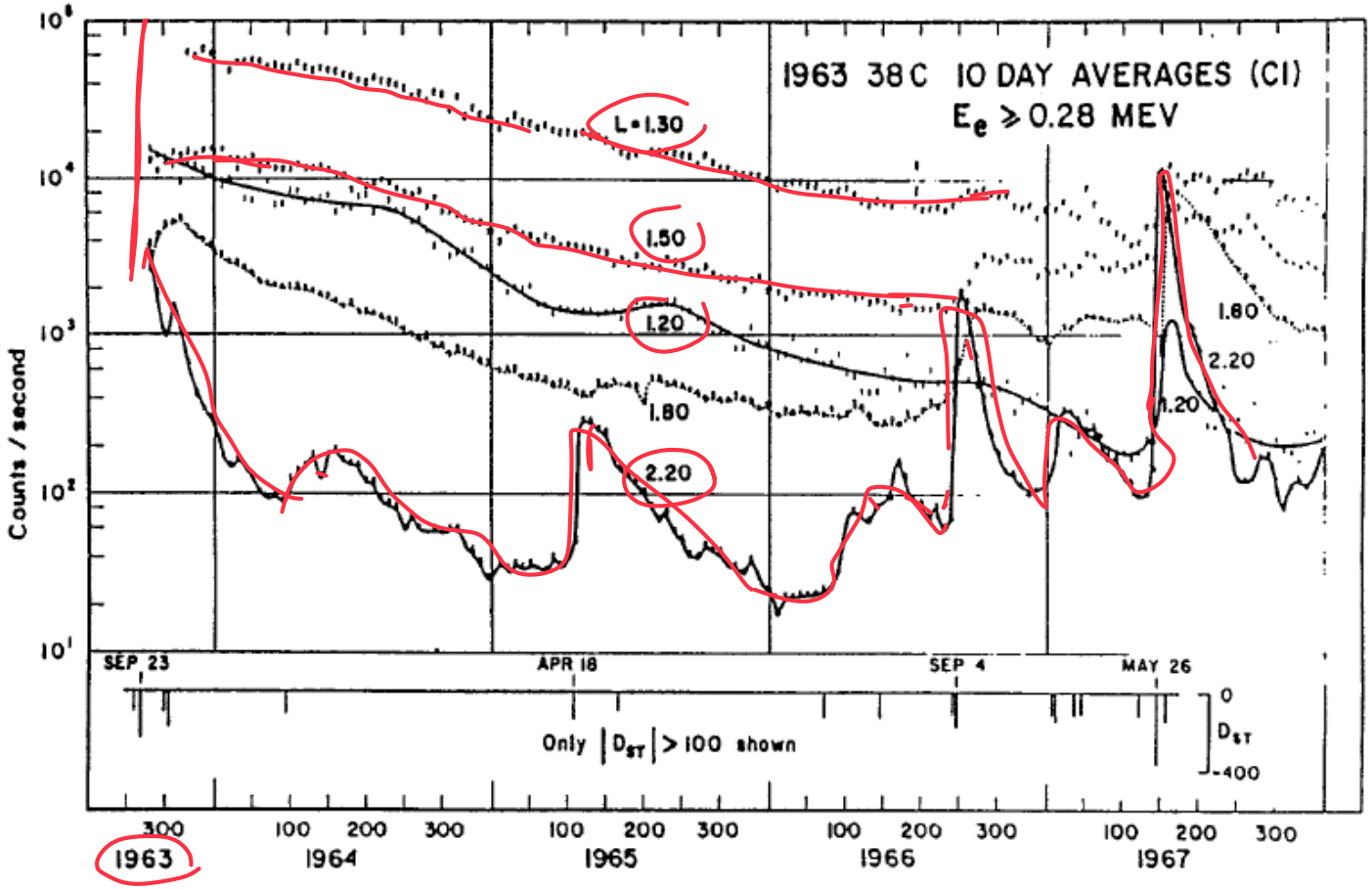
Radiation Belt Models: AE9 and AP9 -> (IRENE: International Radiation Environment Near-Earth)

- AE8 / AP8 models are NASA models developed in 1991
  - simple MAX / MIN conditions; no other time variation
- AE9 / AP9 finally developed by AFRL and released in 2012; regularly updated since
  - based on >37 satellite data sets results given as statistical confidence intervals



Starfish Prime

- Artificial radiation belts created by nuclear detonation at 400 km altitude above Johnston Island in the South Pacific: July 9, 1962
- Telstar was launched one day after the detonation…
  failed due to a 100x higher radiation exposure than expected



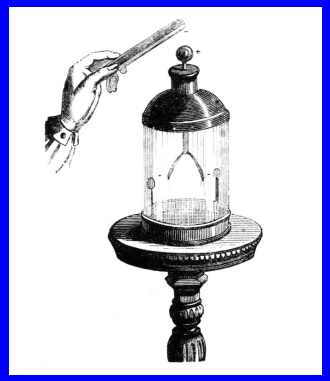
10 day averages of the inner zone electron fluxes greater than 0.28 MeV (in units of count rate) measured by the satellite 1963-38C. The effects of the decay of the Starfish nuclear debris and of the 4 major geomagnetic perturbations are evident.[51]

University of Colorado **Boulder**

Radiation Belt Dynamics + Sources

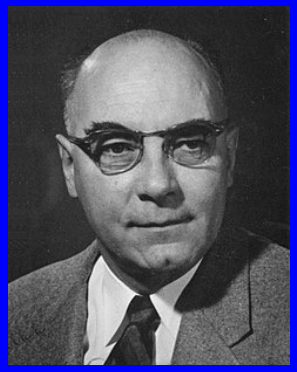
Xinlin Li

In early 1900s scientists had been puzzled by the levels of ionizing radiation measured in the atmosphere and on Earth

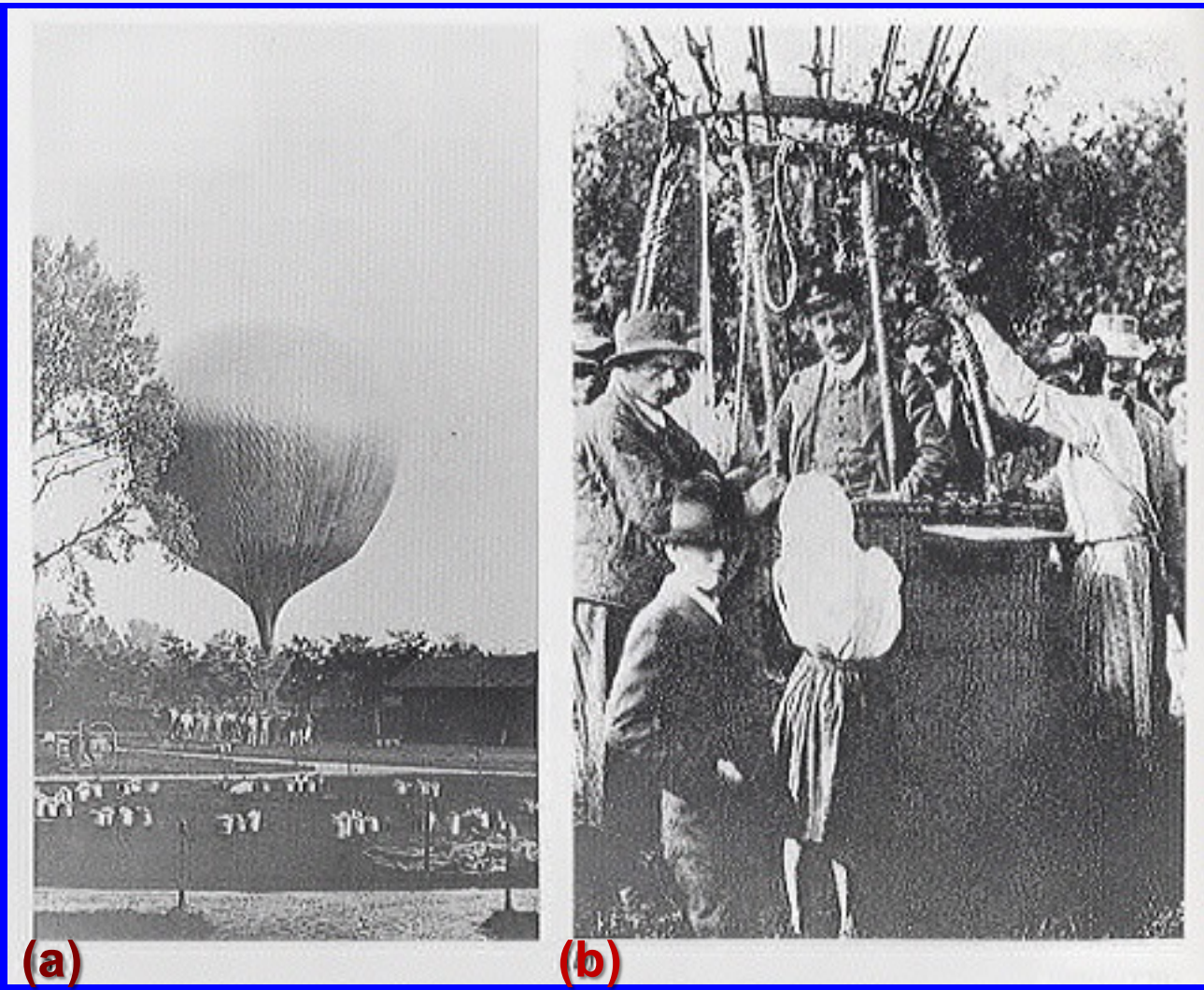


R. A. Millikan

Millikan proved that this radiation is indeed of extraterrestrial origin, and named it "cosmic rays" in 1925.



C. D. Anderson



(a) (b)

The balloon flights of Victor Hess
(a) Preparation of one of his flights in 1911-12
(b) Hess after a successful Balloon flight

Positron and muon were discovered in cosmic rays by Anderson.

Hess and Anderson shared the 1936 Nobel Prize in Physics